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Unpacking Earth's Energy Imbalance: Cumulative Shifts in Climate Health

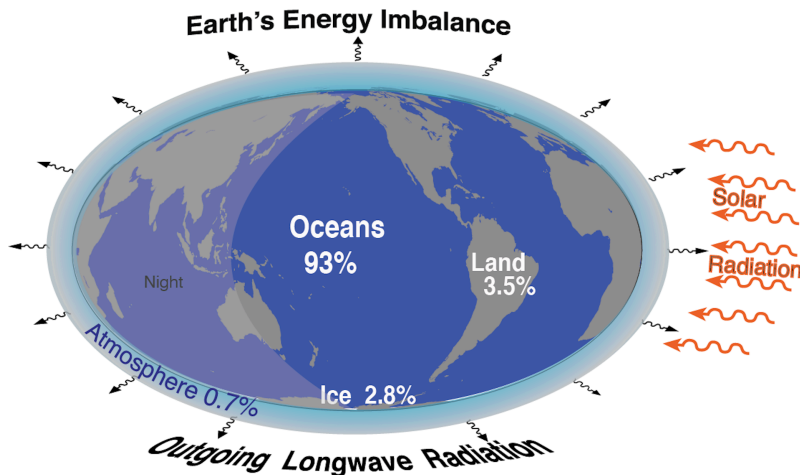
Abstract

Climate change can sound small: another tenth of a degree, another centimeter of sea level. But in 2021, climate scientist Kevin Trenberth explained that these small changes accumulate relentlessly. Earth is trapping about 460 terawatts of extra energy, far more than all the energy humans use. This lesson follows that energy into the oceans, ice, land, and atmosphere, and explains why risks rise faster than a straight line.

Introduction

July 2021 was Earth's hottest month on record, marked by extreme storms, floods, and wildfires. That year, the Intergovernmental Panel on Climate Change stated more emphatically than ever that human activities are damaging the climate and that effects are rapidly getting worse. Trenberth's explanation begins with energy. The Sun delivers about 173,600 terawatts of energy to Earth. About 30% is reflected back to space, leaving 122,100 terawatts to drive weather, climate, and the water cycle. Almost all of that eventually radiates back to space, except for about 460 terawatts trapped by greenhouse gases. That leftover energy is Earth's energy imbalance, or in other words, global warming.

How Do Scientists Study This?

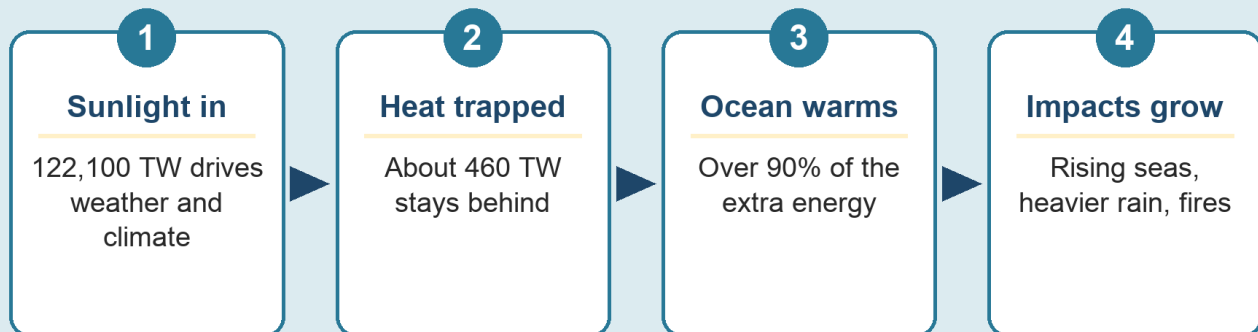


FROM THE ARTICLE

Earth's energy imbalance of 460 terawatts and the share going into oceans, land, ice, and atmosphere. Kevin Trenberth, CC BY-ND.

Scientists quantify Earth's energy budget by measuring incoming and outgoing radiation and tracking where extra energy accumulates. They measure carbon dioxide concentrations, which Dave Keeling began recording at Mauna Loa, Hawaii, in 1958. They track ocean heat content at different depths from 1958 onward, sea level from aircraft and satellites since 1992, and Arctic sea ice since 1979. By adding up the energy going into the ocean, ice, land, and atmosphere, they can account for the whole imbalance. They also compare it with human energy use: world electricity generation in 2018 was about 2.6 terawatts, and all energy use about 19.5 terawatts.

Following Earth's extra energy



Study the Process

Balance the budget

Measure incoming sunlight, reflected light, and outgoing heat to find what is trapped.

Follow the heat

Track ocean heat by depth, sea level, sea ice, and land temperatures over decades.

Compare with people

Put the 460 TW imbalance next to humanity's total energy use.

What Did They Discover?

Carbon dioxide rose from about 280 ppm before industrialization to 310 ppm in 1958 and about 415 ppm today, a 48% increase. Over 90% of the extra energy goes into the oceans. The upper ocean began warming around the 1970s, heat reached 500 to 1,000 meters by the early 1990s, and below 1,500 meters by 2005. Sea level rose about 3 mm per year from 1992 to 2012 and about 4 mm since, over 90 mm (3.5 inches) in 29 years, causing sunny-day flooding in places like Miami. About 13 TW melts ice; summer Arctic sea ice has shrunk over 40% since 1979. Air over oceans is 5% to 15% moister than before 1970, boosting heavy rain by about 10%, and sometimes 30% as storms intensify.

The imbalance by the numbers

460 TW extra energy trapped by greenhouse gases

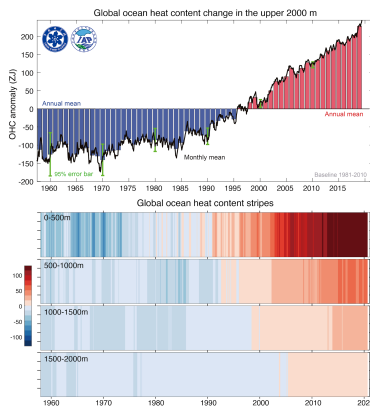
19.5 TW all energy humans use worldwide, for comparison

48% rise in carbon dioxide above pre-industrial levels

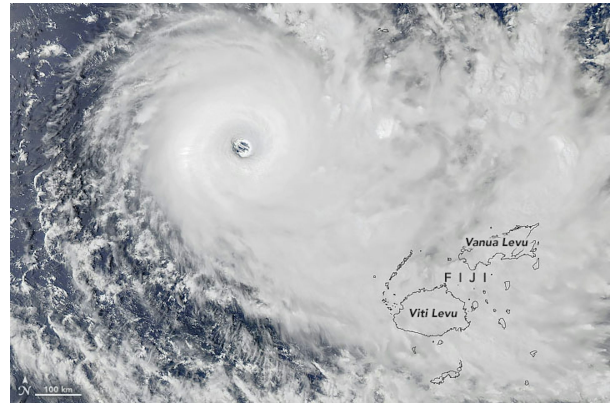
Trenberth calls it the straw that breaks the camel's back: risks rise nonlinearly, much faster than a straight line, as thresholds are crossed.

Look at the Evidence

FROM THE ARTICLE



Ocean heat content change in the upper 2,000 meters, 1958-2020, and warming by depth. Cheng et al. (2021), CC BY-ND.



Cyclone Yasa heads for Fiji in December 2020, the fourth most intense tropical cyclone on record in the South Pacific. Image: NASA Earth Observatory.

Takeaways

Small but huge

The 460 TW imbalance is under 1% of natural flows but dwarfs human energy use.

Heat accumulates

Most extra energy builds up in the ocean, raising seas and fueling storms.

Risks accelerate

Crossing thresholds makes damages rise much faster than warming itself.

Key Terms

Terawatt (TW)

One trillion watts, a unit used to describe very large flows of energy.

Energy imbalance

The extra energy Earth absorbs but does not return to space, which heats the planet.

Evapotranspiration

Water moving from land to air through evaporation and plants releasing water vapor.

Sunny-day flooding

Coastal flooding during high tides even without a storm, caused by rising sea levels.

Nonlinear

Changing in a way that is not a straight line, so effects can speed up suddenly.

Match-Up: draw a line from each term to its clue

A Terawatt (TW)

B Energy imbalance

C Evapotranspiration

D Sunny-day flooding

E Nonlinear

1 Changing in a way that is not a straight line, so effects...

2 Water moving from land to air through evaporation and plants releasing water...

3 The extra energy Earth absorbs but does not return to space, which...

4 One trillion watts, a unit used to describe very large flows of...

5 Coastal flooding during high tides even without a storm, caused by rising...

Answer key: A-4 · B-3 · C-2 · D-5 · E-1

Activity

Where Does the Energy Go?

Students build a visual budget showing how Earth's extra energy is divided among oceans, ice, land, and air.

Materials: Graph paper or a 10×10 grid (100 squares); Colored pencils; Calculator

1. Shade 93 squares for the ocean, about 3 for ice, about 3 for land, and 1 for the atmosphere.
2. Label each section with one consequence from the lesson, such as sea level rise or drought.
3. Compare 460 TW with world electricity (2.6 TW) and all energy use (19.5 TW). How many times larger is the imbalance?
4. Calculate sea level rise over 29 years at 3 mm and at 4 mm per year.
5. Explain in writing why small yearly changes can still lead to big consequences.

References

Trenberth, K. (2021). Climate change is relentless: Seemingly small shifts have big consequences. The Conversation. <https://doi.org/10.64628/AAI.6nxhm4fga>